

EDUCATION

- **BTech Electrical Engineering**, National Institute of Technology, Jaipur 2025.

EXPERIENCE

ICICI Bank (July 2025 – Present) | Data Engineer, Mumbai

- Worked with **IBM InfoSphere DataStage and Designer** to schedule and monitor automated ETL jobs, ensuring reliable batch processing and timely data availability across enterprise systems.
- Analyzed and resolved **UCIC (Unique Customer Identification Code) mismatches** by investigating MDM deduplication rules and API matching logic, improving customer identity accuracy across downstream applications.
- Built and enhanced **Azure ETL pipelines** using **Azure Data Factory and Azure Databricks** to ingest and transform source-system data, delivering analytics-ready datasets to business and risk teams.
- Implemented **PII and PCI data redaction** using **regular expressions** to identify sensitive fields across **on-premise Oracle and MySQL databases**, strengthening data security and compliance controls.

PROJECTS

Multi-Agent Content Intelligence & Publishing System

- Designed a multi-agent system using **LangGraph** where specialised agents (research agent, drafting agent, quality-check agent) coordinate via shared state to autonomously generate and publish LinkedIn content.
- Exposed agent orchestration logic through a **FastAPI** microservice, enabling async job submission, streaming LLM responses, and webhook-based triggering from the n8n workflow layer.
- Containerised the full stack with **Docker** on **GCP**, managing session persistence, secret injection, and retry logic for production-grade reliability.

Conversational AI Assistant with Context Memory (WhatsApp)

- Built a production conversational AI assistant on WhatsApp using WAHA + **LangChain**, implementing session-based context management and a short-term memory buffer to maintain conversation continuity.
- Designed a vector-backed retrieval layer (**ChromaDB**) for FAQ grounding, reducing hallucinated responses on domain-specific queries.
- Deployed async event-handling backend with FastAPI and Docker on a GCP Linux VM, integrating a rule-based AI classifier to filter unsafe or off-topic responses before delivery.

Agentic RAG Pipeline for Quantitative Alpha Research

- Built **RAG pipeline** using LangGraph to orchestrate an autonomous research agent that ingests unstructured financial documents, retrieves relevant context via semantic search, and generates structured alpha signal using **LLMs**.
- Built a vector retrieval layer using **ChromaDB** with **Gemini embeddings** and cosine similarity search, grounding LLM outputs to reduce hallucination on domain-specific financial data..
- Implemented async Python workflows with queue-based parallel execution and SQL logging for experiment tracking. Used World quant Brain api endpoint to programmatically submit alpha formulas and retrieve backtesting results.

TECHNICAL SKILLS

- Programming language: Python, SQL
- AI & ML: Pytorch, TensorFlow, LSTM, NLP, LangChain, LangGraph, Hugging Face, ChromaDB
- Database & Analytics Tools: ETL (IBM DataStage), MDM, Azure data factory, Databricks.
- Data visualisation tools: Tableau, Power Bi
- Software: Visual Studio Code, GitHub Desktop, MS Office (Excel, Word, PowerPoint), GCP
- Others: Dynatrace, Postman, Spark SQL, SQL Developer, n8n, Docker

LEADERSHIP SKILLS

- **Executive, Think India MNIT Club:** Led the planning and execution of large-scale events with **5,000+ attendees**, managing and coordinating a team of **20+ students** to ensure smooth operations and delivery.
- **Founder, Unicorn Club (2022):** Built and scaled a community of entrepreneurs, organizing and hosting **30+ talks** focused on practical business insights, collaboration, and knowledge sharing.